

ECG Anomaly Detection — Session Report

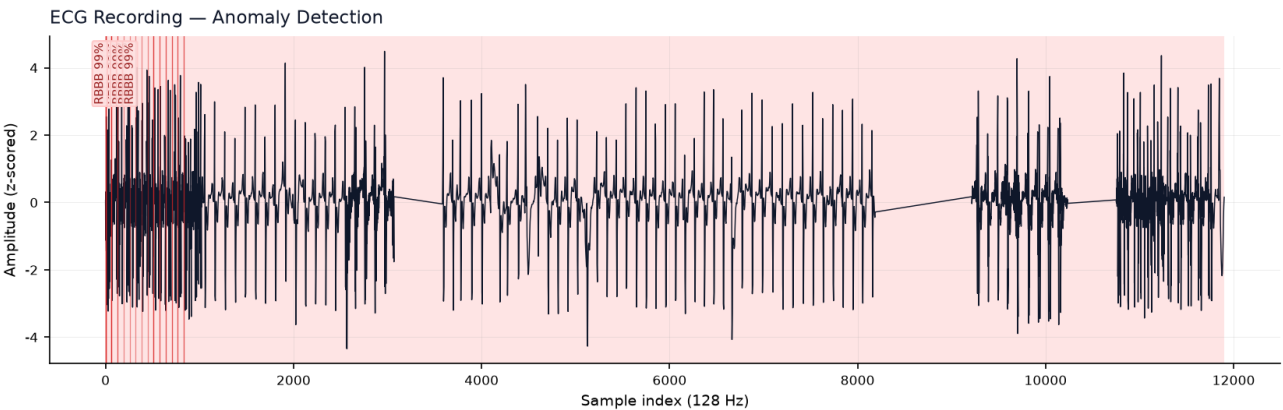
Prepared for: op • Session ID: 77f70b83 • Generated: 2026-07-05 06:56:30

Session Information

Patient / User	Sriyantha Sadaruwan
Source	mit-bih-arrhythmia (118)
Started	2026-07-05 06:56:30
Ended	2026-07-05 06:58:26
Duration	113.6 s
Status	completed
Total beats analyzed	56,832
Anomalous beats	37,947 (66.77%)
Model	ecg-ae-v2 (v1783189189) • Dilated-U-Net-Autoencoder
Detection threshold (MAE)	0.1341

ECG Recording & Detection Results

The following plot shows the recorded ECG signal (z-score normalized). Red-shaded regions indicate samples flagged as anomalous by the autoencoder (reconstruction error exceeding the trained threshold).



Main Arrhythmia Finding

Right Bundle Branch Block was the main high-confidence arrhythmia type detected in this session. It appeared 574 time(s) with average confidence 98.9% and maximum confidence 99.2%.

Reason: this type is shown because the autoencoder first flagged the ECG segment as anomalous by reconstruction error, and the classifier then assigned this arrhythmia label with at least 50% confidence. Lower-confidence labels are excluded from this report to avoid over-interpreting uncertain model output.

Example: time 2026-07-05 06:57:55, anomaly score 0.1253, threshold 0.0500, classifier confidence 99.2%.

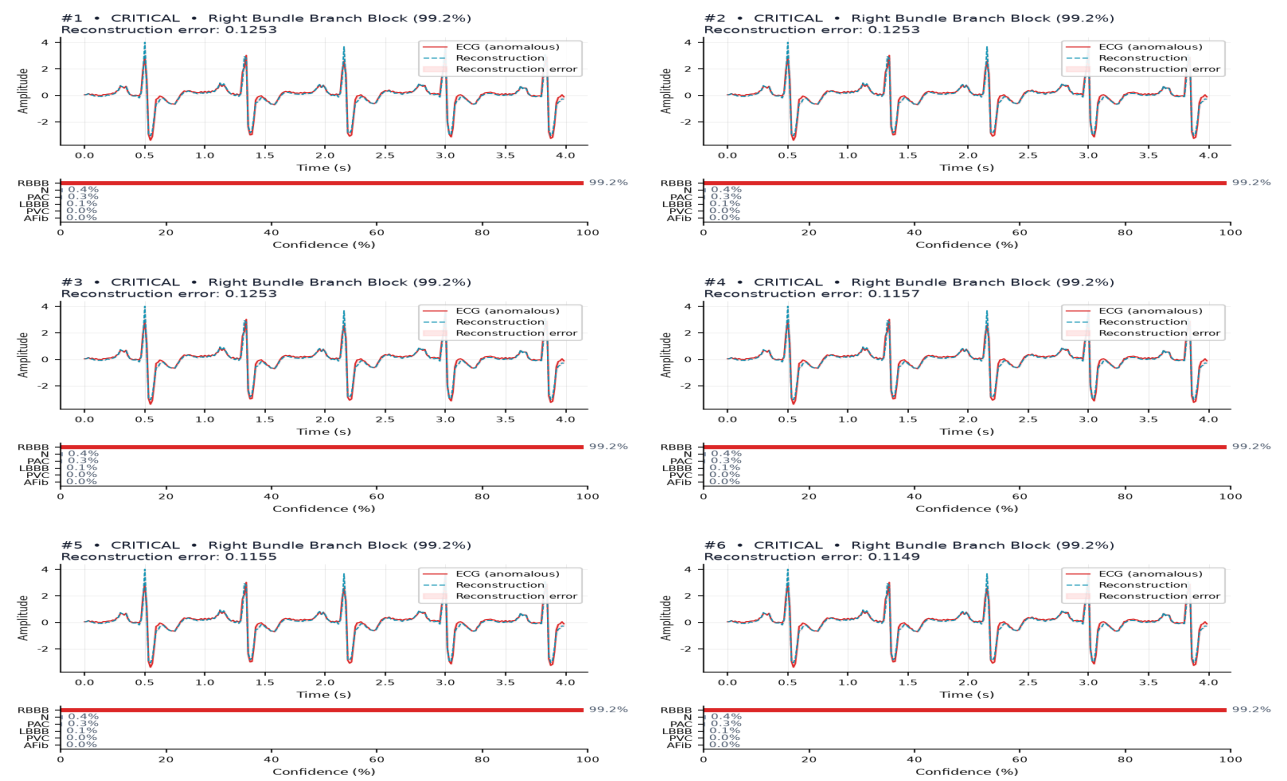
Clinical meaning: Right Bundle Branch Block (RBBB) is a condition where electrical signals to the right ventricle are delayed, causing it to contract later than the left ventricle. This produces a characteristic ECG pattern with a wide QRS complex. RBBB is generally less concerning than LBBB and can occur in otherwise healthy individuals.

Why it matters: RBBB in the absence of other heart disease is usually benign and does not require treatment. However, new RBBB combined with symptoms like chest pain or shortness of breath should be evaluated by a cardiologist to rule out underlying conditions.

Arrhythmia Type	Detections	Avg Confidence	Example
Right Bundle Branch Block	574	98.9%	score 0.1253, conf 99.2%
Premature Atrial Contraction	70	90.0%	score 0.1137, conf 98.3%

Detected Anomaly Waveforms

The following plots show example ECG segments (red) that triggered alerts, alongside the autoencoder's reconstruction of what it expected to see (blue dashed). The shaded region between the two curves represents the reconstruction error that exceeded the detection threshold. When available, examples are selected from arrhythmia classifications above 50% confidence.



Clinical Summary

66.8% of analyzed samples were flagged as anomalous, suggesting sustained abnormal ECG morphology. Urgent cardiology review is recommended. High-confidence classifier findings in this report include: Right Bundle Branch Block, Premature Atrial Contraction. Only arrhythmia labels with confidence above 50% are included.

Physician Notes

Physician signature: _____

Date: _____

This report is generated automatically by an unsupervised deep-learning anomaly detection system and is intended for decision support only. It does not constitute a medical diagnosis and should not replace clinical judgment by a qualified physician.